

Hidden-Package Incorrect-Answer Audit

Every mismatch from the Qwen3-VL Plus Gear Assemblies hidden evaluation

Field	Value
Model	qwen3-vl-plus
Evaluation date	2026-08-11
Repository commit	360772e84d93f63010c806fc0faf21654bb8c4fc
Scope	28 source-hidden packages; hidden answer key kept local

Purpose: document methodology, exact-answer performance, question-type error rates, token use, estimated list-price cost, and per-item failures.

Prepared from retained provider responses, blind grading outputs, and independent reconciliation checks.

Audit overview

This companion report enumerates all 20 incorrect hidden-package answers. Every record includes the original question, model prediction, correct answer, signed numeric error, and a concise descriptive diagnosis. Signed error equals prediction minus correct answer. Negative values are underestimates, and positive values are overestimates.

All 20 errors were numeric. Fifteen were underestimates and five were overestimates. Transmission ratios contributed seven errors, tooth counts contributed 11, and component counts contributed two. The gear-type item was correct, and no axis-angle item existed in the hidden set.

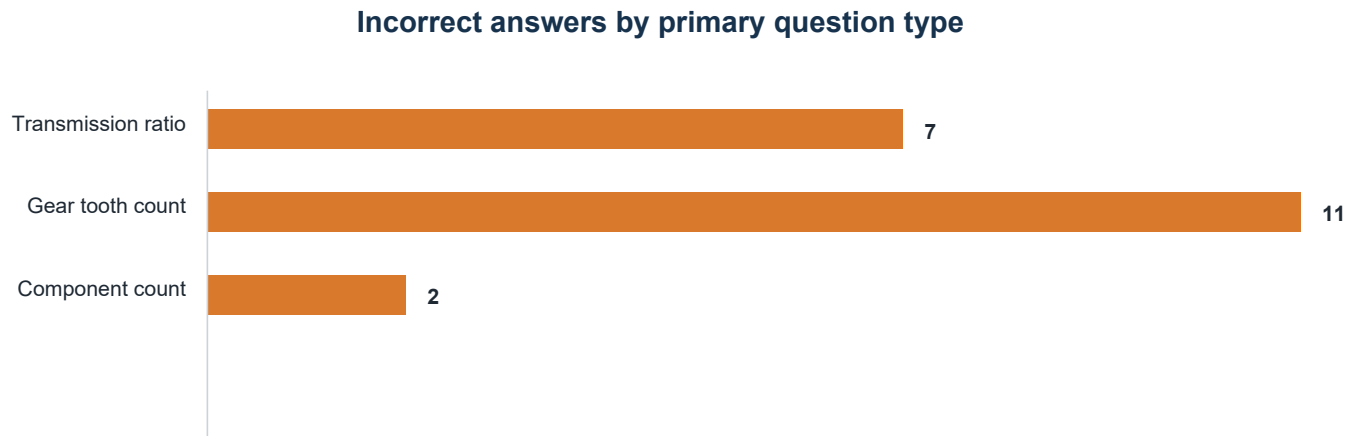


Figure 1. Incorrect hidden-answer counts by primary question type.

Transmission ratio

Package	Subtype	Question	Predicted	Correct	Signed error	Diagnostic
Model02_01_Hidden	Other train or gearbox ratio	what is the transmission ratio of the gear train?	12	16	-4.00; -25.0%	The transmission ratio was underestimated by 4 (25.0% of the correct value).
Model03_01_Hidden	Planetary ratio	what is the transmission ratio of this planetary gear train? Assume the sun gear is the input, ring gear is fixed. And carrier is the output.	3	6	-3.00; -50.0%	The transmission ratio was underestimated by 3 (50.0% of the correct value).
Model04_01_Hidden	Planetary ratio	what is the transmission ratio of this planetary gear? Assume sun input, ring fixed, carrier output. Round up to the nearest integer.	4	5	-1.00; -20.0%	The transmission ratio was underestimated by 1 (20.0% of the correct value).
Model06_04_Hidden	Other train or gearbox ratio	What is the transmission ratio of this gear box?	3	2	+1.00; +50.0%	The transmission ratio was overestimated by 1 (50.0% of the correct value).
Model08_03_Hidden	Other train or gearbox ratio	What the transmission ratio of this gear box? Round to the first decimal place.	5.0	3.6	+1.40; +38.9%	The transmission ratio was overestimated by 1.4 (38.9% of the correct value).
Model09_02_Hidden	Other train or gearbox ratio	What is the transmission ratio of the gear box? Round to the first decimal place.	3.0	5.4	-2.40; -44.4%	The transmission ratio was underestimated by 2.4 (44.4% of the correct value).
Model10_01_Hidden	Other train or gearbox ratio	what is the transmission ratio of this gear train?	1	32	-31.00; -96.9%	The transmission ratio was underestimated by 31 (96.9% of the correct value).

Gear tooth count

Package	Subtype	Question	Predicted	Correct	Signed error	Diagnostic
Model03_02_Hidden	Ring gear teeth	What is the tooth number of the ring gear?	72	80	-8.00; -10.0%	The tooth count was underestimated by 8 (10.0% of the correct value).
Model03_03_Hidden	Sun gear teeth	What is tooth number of the sun gear?	12	20	-8.00; -40.0%	The tooth count was underestimated by 8 (40.0% of the correct value).
Model04_02_Hidden	Ring gear teeth	What is the tooth number of the ring gear?	48	65	-17.00; -26.2%	The tooth count was underestimated by 17 (26.2% of the correct value).
Model04_03_Hidden	Sun gear teeth	What is tooth number of the sun gear?	24	16	+8.00; +50.0%	The tooth count was overestimated by 8 (50.0% of the correct value).
Model04_04_Hidden	Planet gear teeth	What is the tooth number of planet gear?	24	34	-10.00; -29.4%	The tooth count was underestimated by 10 (29.4% of the correct value).
Model06_02_Hidden	Smaller gear teeth	What the tooth number of the smaller gear?	12	25	-13.00; -52.0%	The tooth count was underestimated by 13 (52.0% of the correct value).
Model06_03_Hidden	Larger gear teeth	What is the tooth number of the larger gear?	30	50	-20.00; -40.0%	The tooth count was underestimated by 20 (40.0% of the correct value).
Model08_01_Hidden	Smaller gear teeth	what the tooth number of the smaller gear?	12	29	-17.00; -58.6%	The tooth count was underestimated by 17 (58.6% of the correct value).
Model08_02_Hidden	Larger gear teeth	what the tooth number of the larger gear?	60	105	-45.00; -42.9%	The tooth count was underestimated by 45 (42.9% of the correct value).
Model09_01_Hidden	Sun gear teeth	What is the tooth number of the sun gear?	24	17	+7.00; +41.2%	The tooth count was overestimated by 7 (41.2% of the correct value).
Model09_03_Hidden	Ring gear teeth	What is the tooth number of the ring gear?	48	75	-27.00; -36.0%	The tooth count was underestimated by 27 (36.0% of the correct value).

Component count

Package	Subtype	Question	Predicted	Correct	Signed error	Diagnostic
Model06_01_Hidden	Assembly inventory count	How many bearings are there is in this gearbox?	4	2	+2.00; +100.0%	The component count was overestimated by 2 (100.0% of the correct value).
Model07_01_Hidden	Assembly inventory count	How many gears are there in this gear box?	3	4	-1.00; -25.0%	The component count was underestimated by 1 (25.0% of the correct value).

Correct-answer control cases

These eight items were correct and account for the remainder of the 28-question denominator.

Package	Type	Question	Correct answer	Prediction
Model02_02_Hidden	Component count	How many support brackets are there in this assembly?	0	0
Model02_03_Hidden	Component count	How many distinct parts are there in this assembly?	4	4
Model03_04_Hidden	Component count	How many planet gears are there in this planetary gear train?	3	3
Model04_05_Hidden	Component count	How many planet gears are there in this planetary gear train?	3	3
Model07_02_Hidden	Component count	How many shafts are there in this gear box?	3	3
Model07_03_Hidden	Transmission ratio	What is the transmission ratio of this gear train?	12	12
Model08_04_Hidden	Gear type	What is the type of the gear shape? Spur, Helical, Bevel or Double Helical or others.	Helical	Helical
Model09_04_Hidden	Component count	How many planet gears are there?	4	4

Error-pattern synthesis

Gear-tooth enumeration remained the dominant failure mode: every ring, sun, planet, smaller-gear, and larger-gear target was wrong. Ratio answers also usually fell below the correct value. In contrast, only two component inventory questions were wrong, and all planet-gear quantity items were correct.

Interpretation limits

The diagnostics describe observed answer differences and do not establish internal model mechanisms. Direct-answer generation suppressed reasoning, so this audit cannot distinguish visibility loss, cross-view correspondence, tooth enumeration, kinematic setup, arithmetic, and output-selection errors.

References

[1] Alibaba Cloud. Qwen-VL compatible with OpenAI. Accessed Aug. 11, 2026. <https://www.alibabacloud.com/help/en/model-studio/qwen-vl-compatible-with-openai>

[2] axioforgeai. gear-assemblies-qa. GitHub commit 360772e84d93f63010c806fc0faf21654bb8c4fc. <https://github.com/axioforgeai/gear-assemblies-qa/tree/360772e84d93f63010c806fc0faf21654bb8c4fc>